

RAYEN NACEF | AI ENGINEER

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PROFESSIONAL SUMMARY

AI Engineer with experience across the full ML lifecycle — from data pipelines and fine-tuning to RAG systems and production deployment. Research background in LLM optimization and RL. Actively seeking a professional position to apply and grow my skills in a real-world engineering environment.

PROFESSIONAL EXPERIENCE

Data Scientist | Ominimo Insurance Group — Serbia

Feb – May 2026

- Engineered an end-to-end data validation pipeline with automated quality checks, enforcing domain-specific thresholds on daily ingested data and triggering real-time alerts to the IT team on anomaly detection.
- Designed and deployed a scalable web-crawling pipeline using Apache Airflow and Docker, automating structured data collection and feeding clean datasets into downstream analytics workflows. improving pipeline reliability and data integrity for insurance analytics.

Keywords: Data Validation, Data Quality Assurance, Web Crawling, Apache Airflow, Python, Docker, Pipeline Automation

AI Research Intern - MITACS Program | Mila & Polytechnique Montréal — Canada

Jul – Sep 2025

- Optimized Open-R1 LLM training infrastructure, reducing GPU requirements from 8 to 1 while preserving model performance through memory-efficient techniques and gradient checkpointing.
- Designed a novel token-level credit assignment mechanism for Group Relative Policy Optimization (GRPO), achieving a 13% improvement in mathematical reasoning accuracy on benchmark evaluations.

Keywords: LLM, Reinforcement Learning (RL), GRPO, Fine-Tuning, PyTorch, Transformers, HPC, Slurm, Linux Bash, Python, Reasoning, GitHub

AI Engineer Intern | Proxiad — Tunis, Tunisia [GitHub]

Feb – May 2025

- Architected a production-ready RAG-based chatbot by orchestrating two LLMs, integrating a vector knowledge base with structured data preprocessing for accurate retrieval.
- Deployed the chatbot system for real-time e-commerce order tracking and claims handling using FastAPI, containerized with Docker and Kubernetes for scalable, fault-tolerant production performance.

Keywords: LLM, RAG, Prompt Engineering, FastAPI, Data Preprocessing, Data Validation, Python, Java, Docker, Kubernetes, NLP, SpringBoot, Angular

AI Engineer Intern | Focus Corporation — Tunisia

Jul – Aug 2024

- Built a modular data preprocessing pipeline to clean, transform, validate, and analyze both structured and unstructured datasets, preparing high-quality training corpora for downstream LLM fine-tuning.
- Developed an AI-powered network diagnostic assistant capable of parsing Wireshark packet capture files and generating natural-language root-cause analysis, reducing manual inspection time.

Keywords: LLM, RAG, Pandas, Data Cleaning, Prompt Engineering, Python, Hugging Face, Docker, Ollama, NLP

SELECTED PROJECTS

CryptoFacial Recognition System | Embedded AI & Security

Dec 2024 – Jan 2025

- Built a real-time facial recognition system with OpenCV and deep CNN-based face embeddings, deployed on an ESP32 edge device.

- Secured client-device communication using AES encryption over TCP sockets, ensuring data privacy for biometric data transmission.

Keywords: Python, OpenCV, CNN, Face Recognition, AES Encryption, ESP32, TCP Socket, Deep Learning

FreeRTOS-Based Real-Time Robot Control | Embedded Systems

Oct – Nov 2024

- Designed a multitask FreeRTOS control architecture for a robot, synchronizing motor control, sensor fusion, and Wi-Fi communication tasks.
- Implemented a TCP server for remote real-time motor command dispatch and battery telemetry monitoring.

Keywords: C++, FreeRTOS, ESP32, TCP/IP, Multitasking, GPIO, ADC, Embedded Systems

Intelligent Intrusion Detection for CPS [GitHub] | ML & Cybersecurity

May – Jul 2024

- Benchmarked four ML classification algorithms for real-time network intrusion detection in Cyber-Physical Systems (CPS).
- Designed a dynamic algorithm selection module that reduced false-positive rate by 30% compared to single-model baselines.

Keywords: Machine Learning, Scikit-learn, TensorFlow, Keras, Python, CPS Security

HexaSwift Autonomous Hexapod Robot | Robotics & Computer Vision

Feb – Apr 2024

- Engineered an autonomous hexapod robot with ArUco marker-based indoor navigation using OpenCV and PID controllers on Raspberry Pi.
- Developed a companion Django mobile interface for real-time robot teleoperation and sensor feedback visualization.

Keywords: Python, OpenCV, Raspberry Pi, Embedded C, PID Control, ArUco Markers, Arduino, UART, Django

EDUCATION

Computer Science Engineering Diploma — ENSI

2022 – 2025

DEUPC — Preparatory Cycle for Engineering (Physics & Chemistry) — IPEIT

2020 – 2022

TECHNICAL SKILLS

Programming	Python, C, C++, Java
AI / ML / DL	Machine Learning, Deep Learning, Computer Vision, NLP, Reinforcement Learning (RL), LLM, GenAI
GenAI & LLMs	LangChain, LangGraph, Hugging Face Transformers, Ollama, LlamaIndex, RAG, Prompt Engineering, Fine-Tuning, RLHF
MLOps & DevOps	Docker, Kubernetes, CI/CD, GitLab, Minikube, Slurm, HPC, Linux Bash
Frameworks	FastAPI, Flask, Spring Boot, Angular, PyTorch, TensorFlow, Keras, Pandas, Scikit-learn
Databases	PostgreSQL, SQL, Redis
Tools	Git, GitHub, VS Code, Keil, IntelliJ IDEA
Soft Skills	Research Writing, Time Management, Teamwork, Fast Learner
Languages	English, French, Arabic